HUNT-SAGE · THREAT HUNTING CHEAT SHEET

LOLBins Quick Reference

Commonly abused Windows binaries: what they are used for and which arguments give them away.

github.com/iamjammie/hunt-sage · iamjammie.github.io/hunt-sage · Built 2026-10-31

Full catalogue: **LOLBAS**. Linux: **GTFOBins**. Drivers: **LOLDrivers**. macOS: **LOOBins**.

Detection principle: **binary + arguments + parent + network + user**. The binary alone is rarely enough.

Binary	Abuse	Suspicious argument patterns	MITRE
`certutil.exe`	Download, encode/decode	`-urlcache -split -f http...`, `-decode`, `-encode`, `-verifyctl`	T1105, T1140
`bitsadmin.exe`	Download, persistence	`/transfer`, `/addfile`, `/SetNotifyCmdLine`	T1197
`mshta.exe`	Run HTA/script	`http(s)://`, `javascript:`, `vbscript:`	T1218.005
`rundll32.exe`	Run DLL/script	`javascript:`, `comsvcs.dll, MiniDump`, DLL in `Temp` / `AppData`, no-arg spawn	T1218.011, T1003.001
`regsvr32.exe`	Squiblydoo	`/s /n /u /i:http://... scrobj.dll`	T1218.010
`msiexec.exe`	Run remote MSI	`/q /i http://...`	T1218.007
`wmic.exe`	Remote exec, recon, XSL	`/node:`, `process call create`, `/format:http...`, `shadowcopy delete`	T1047, T1220
`powershell.exe`	Everything	`-enc`, `-nop -w hidden`, `IEX`, `DownloadString`	T1059.001
`cmd.exe`	Command chaining	`/c` with `&`, `^` obfuscation, long cmdlines	T1059.003
`wscript.exe` / `cscript.exe`	Script host	`.js`, `.vbs`, `.wsf` from `AppData` / `Temp` / `Downloads`	T1059.005/.007
`schtasks.exe`	Persistence / remote exec	`/create ... /s <host>`, `/ru SYSTEM`, user-path actions	T1053.005
`sc.exe`	Service creation / tamper	`create ... binPath=`, `config ... start= disabled`, remote `\\host`	T1543.003, T1562.001
`reg.exe`	Persistence, dump hives	`add ...\Run`, `save HKLM\SAM|SYSTEM|SECURITY`, `export`	T1547.001, T1003.002
`net.exe` / `net1.exe`	Recon, accounts	`user /add`, `localgroup administrators /add`, `use \\host\ipc$`, `group "domain admins" /domain`	T1136, T1087, T1021.002
`nltest.exe`	Domain recon	`/dclist:`, `/domain_trusts`	T1482
`dsquery.exe`, `adfind.exe`	AD enumeration	`* -filter`, `-f "(objectcategory=...)"`	T1087.002
`whoami.exe`	Recon	`/all`, `/priv`, `/groups`	T1033
`tasklist.exe` / `taskkill.exe`	Recon / kill AV	`/fi`, `/im defender...`	T1057, T1562
`vssadmin.exe` / `diskshadow.exe` / `wbadmin.exe` / `bcdedit.exe`	Impact / NTDS access	`delete shadows`, `recoveryenabled no`, `create shadow`	T1490, T1003.003
`ntdsutil.exe`	Dump AD DB	`"ac i ntds" "ifm" "create full"`	T1003.003
`esentutl.exe`	Copy locked files	`/y /vss`	T1003.003
`msbuild.exe`	Compile & run inline C#	`.csproj` / `.xml` in user path; network connections from msbuild	T1127.001
`installutil.exe`, `regasm.exe`, `regsvcs.exe`	.NET proxy execution	`/U` with DLL from user path	T1218.004/.009
`cmstp.exe`	INF execution, UAC bypass	`/s /ns` with `.inf` from user path	T1218.003
`forfiles.exe`, `pcalua.exe`, `SyncAppvPublishingServer.exe`	Indirect command execution	`/c cmd`, `-a` path, `-p`	T1202
`control.exe`	CPL execution	`.cpl` from user path	T1218.002

Binary	Abuse	Suspicious argument patterns	MITRE
`odbcconf.exe`	DLL load	`/a {REGSVR dll}`	T1218.008
`expand.exe`, `extrac32.exe`, `makecab.exe`	Copy/compress files	cab to/from UNC, ADS paths	T1105, T1560
`curl.exe`, `tar.exe`, `ssh.exe` (built-in)	Transfer, tunneling	`curl -o`, `ssh -R/-L/-D` tunnels	T1105, T1572
`ftp.exe`	Scripted transfer	`-s:script.txt`	T1048
`findstr.exe`	Download from UNC / search creds	`/V /L ... \\host\share`	T1105
`print.exe`, `xcopy.exe`, `robocopy.exe`	Copy files, staging	Copies to admin shares	T1570, T1074
`dnscmd.exe`	DNS server plug-in DLL	`/config /serverlevelplugindll`	T1543.003
`eventvwr.exe`, `fodhelper.exe`, `computerdefaults.exe`, `sdclt.exe`	UAC bypass via registry hijack	Preceding `HKCU\Software\Classes\...\command` write	T1548.002

Detection helpers

- Use PE `OriginalFileName` to catch renamed copies.
- Check **path**: these live in `System32` / `SysWOW64`; copies in `Users`, `Temp` or `ProgramData` are rarely legit.
- **Parent matters**: Office, browser, mail client, `wmiprvse.exe`, `w3wp.exe`, `services.exe`, `wsmprovhost.exe` as parents of the above are high-signal.
- **Network**: `rundll32`, `regsvr32`, `mshta`, `msbuild`, `certutil` making outbound connections is unusual.
- **Command-line obfuscation**: carets (`c^e^r^t^u^t^i^l`), mixed case, env var substring tricks, `%COMSPEC%`, excessive length, `-` vs `/` vs `–` (en dash) switches. Sigma `|windash` handles switch variants.

Related: LOLBin proxy execution playbook.

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